

Genesis Design System (GDS)

Project Genesis Research Repository

Generated: July 2026

Genesis Design System (GDS)

Version 1.0 Visual Language, Tokens & Components

\textbf{Project}: Project Genesis Agentic Artificial Life & Evolutionary Biology Simulator \textbf{Document Role}: Visual design system colors, typography, animations, components \textbf{Companion Documents}: GPS (Product) GES (Engineering) GCR (Content & Research) \textbf{Status}: Living document extend as new components are built

> "Design is not decoration. It is the architecture of experience."

Part I Design Philosophy

1.1 The Visual Identity of Genesis

Genesis exists at the intersection of \textbf{science} and wilderness.

It is a research tool precise, structured, evidence-based. It is also a window into a world dramatic, alive, unknowable.

The design must hold both truths simultaneously.

The result is a visual language built from three sources:

1. \textbf{Space} and deep time. Dark backgrounds. Sparse light. Long durations. The feeling of watching something enormous from a distance.
1. \textbf{Bioluminescence}. Teal accents. Green-to-teal gradients. The visual register of life in dark environments organisms that glow because they have to.
1. \textbf{Scientific instruments}. Monospace data. Grid-aligned layouts. Precise labeling. The aesthetic of a spectrometer readout or a seismograph.

These three sources produce a visual language that is simultaneously beautiful and rigorous.

1.2 One Mode Only

Genesis is \textbf{dark mode} only.

Not because dark mode is fashionable.

Because you do not look at stars in daylight. Because laboratory instruments are dark. Because the world in Genesis is rendered on a dark canvas. The light elements the teal rivers, the amber colony, the red death events only sing against darkness.

There is no light mode. There is no toggle.

Part II Color System

2.1 Background Stack

```
/* These four layers create the depth hierarchy of the interface */

--bg-void:      hsl(222, 30%, 4%);      /* #070A10      absolute dark,
page background */
--bg-primary:    hsl(222, 28%, 7%);      /* #0C1019      primary surface */
--bg-secondary: hsl(222, 24%, 10%);      /* #141824      slightly elevated
cards */
```

```
--bg-surface:    hsl(222, 20%, 14%);    /* #1C2130    elevated elements
(inputs, etc.) */
--bg-elevated:   hsl(222, 18%, 18%);    /* #242C3D    highest elevation
(modals) */
```

\textbfRule: Each surface level is exactly one stop above the previous. Never skip levels.

2.2 Accent Bioluminescent Teal

```
--accent-50:    hsl(172, 80%, 95%);    /* Near white teal    for text on
teal bg */
--accent-100:   hsl(172, 75%, 88%);
--accent-200:   hsl(172, 75%, 75%);
--accent-300:   hsl(172, 75%, 62%);
--accent-400:   hsl(172, 80%, 52%);    /* Main accent    use for CTAs,
active states */
--accent-500:   hsl(172, 80%, 45%);    /* Hover state */
--accent-600:   hsl(172, 70%, 36%);    /* Pressed state */
--accent-700:   hsl(172, 60%, 28%);    /* Borders, dim accents */
--accent-800:   hsl(172, 50%, 18%);    /* Subtle accent backgrounds */
--accent-900:   hsl(172, 40%, 10%);    /* Very subtle accent tint */

--accent-glow-sm: 0 0 12px hsl(172, 80%, 48%, 0.20);
--accent-glow-md: 0 0 24px hsl(172, 80%, 48%, 0.25);
--accent-glow-lg: 0 0 48px hsl(172, 80%, 48%, 0.30);
```

2.3 Colony Colors

These match the simulation's colony color system exactly. Never change them they must be consistent between the visualizer and the website.

```
--colony-alpha:    hsl(210, 85%, 58%);    /* Cobalt Blue */
--colony-alpha-dim: hsl(210, 60%, 30%);
--colony-alpha-glow: hsl(210, 85%, 58%, 0.20);

--colony-beta:     hsl(35, 88%, 58%);    /* Warm Amber */
--colony-beta-dim:  hsl(35, 60%, 30%);
--colony-beta-glow: hsl(35, 88%, 58%, 0.20);

--colony-gamma:    hsl(280, 78%, 63%);    /* Violet */
--colony-gamma-dim: hsl(280, 55%, 30%);
--colony-gamma-glow: hsl(280, 78%, 63%, 0.20);

--colony-delta:    hsl(140, 68%, 52%);    /* Emerald Green */
--colony-delta-dim: hsl(140, 50%, 26%);
--colony-delta-glow: hsl(140, 68%, 52%, 0.20);
```

2.4 Semantic Event Colors

```
--event-birth:    hsl(120, 58%, 52%);    /* Verdant green */
--event-death:    hsl(0, 68%, 52%);    /* Blood red */
--event-milestone: hsl(45, 88%, 58%);    /* Gold */
--event-disaster:  hsl(18, 88%, 52%);    /* Volcanic orange */
--event-dispute:   hsl(300, 58%, 58%);    /* Electric violet */
--event-climate:   hsl(195, 75%, 52%);    /* Ice blue */
```

```
--event-extinction:    hsl(0, 50%, 38%);    /* Dark rust      muted
    for gravity */

/* Dot shadows for event markers in timeline */
--event-birth-shadow:   0 0 6px hsl(120, 58%, 52%, 0.5);
--event-death-shadow:  0 0 6px hsl(0, 68%, 52%, 0.5);
--event-milestone-shadow: 0 0 6px hsl(45, 88%, 58%, 0.5);
```

2.5 Biome Colors

```
/* Match visualizer.html palette exactly */
--biome-ocean:          hsl(207, 75%, 28%);
--biome-glacier:        hsl(200, 35%, 82%);
--biome-tundra:         hsl(200, 22%, 60%);
--biome-taiga:          hsl(142, 38%, 30%);
--biome-temperate:      hsl(118, 48%, 36%);
--biome-grassland:      hsl(78, 58%, 42%);
--biome-desert:         hsl(36, 65%, 50%);
--biome-rainforest:     hsl(152, 68%, 26%);
--biome-lake:           hsl(202, 72%, 48%);
```

2.6 Text Colors

```
--text-primary:        hsl(220, 18%, 92%);    /* Near white      main body
    */
--text-secondary:      hsl(220, 14%, 68%);    /* Muted           secondary info
    */
--text-tertiary:       hsl(220, 10%, 48%);    /* Very muted      labels,
    placeholders */
--text-accent:         hsl(172, 80%, 52%);    /* Teal            links, highlights
    */
--text-inverse:        hsl(222, 28%, 7%);     /* Dark            text on teal
    buttons */
--text-danger:         hsl(0, 70%, 60%);      /* Red             errors */
--text-warning:        hsl(38, 85%, 58%);     /* Amber           warnings */
--text-success:        hsl(142, 65%, 52%);    /* Green           success states
    */
```

2.7 Border Colors

```
--border-subtle:       hsl(222, 20%, 18%);    /* Barely visible
    structural dividers */
--border-default:      hsl(222, 20%, 24%);    /* Default card/input borders
    */
--border-hover:        hsl(222, 20%, 32%);    /* Hover state borders */
--border-accent:       hsl(172, 60%, 32%);    /* Teal borders (active
    states, focus) */
--border-danger:       hsl(0, 60%, 40%);      /* Error state borders */
```

2.8 Gradients

```

/* Primary CTA gradient */
--gradient-cta: linear-gradient(135deg,
  hsl(172, 80%, 42%) 0%,
  hsl(192, 80%, 38%) 100%);

/* Card accent top border gradient (applied as border-top) */
--gradient-card-top: linear-gradient(90deg,
  hsl(172, 80%, 48%) 0%,
  hsl(192, 70%, 45%) 50%,
  transparent 100%);

/* World map temperature gradient */
--gradient-temperature: linear-gradient(90deg,
  hsl(230, 80%, 35%) 0%, /* Cold arctic blue */
  hsl(195, 70%, 50%) 25%, /* Cool cerulean */
  hsl(55, 70%, 52%) 60%, /* Warm amber */
  hsl(15, 80%, 50%) 100%); /* Hot red-orange */

/* World map elevation gradient */
--gradient-elevation: linear-gradient(90deg,
  hsl(207, 75%, 20%) 0%, /* Deep ocean */
  hsl(207, 60%, 35%) 28%, /* Shallow water */
  hsl(118, 40%, 32%) 32%, /* Lowland green */
  hsl(78, 40%, 40%) 50%, /* Mid-elevation */
  hsl(35, 30%, 50%) 70%, /* Highland */
  hsl(0, 0%, 75%) 100%); /* Snow cap */

```

Part III Typography

3.1 Font Stack

```

@import url('https://fonts.googleapis.com/css2?family=Space+Grotesk:
  wght@300;400;500;600;700&family=Inter:wght@400;500;600&family=
  JetBrainsMono:wght@400;500&display=swap');

--font-display: 'Space Grotesk', system-ui, -apple-system, sans-serif;
--font-body: 'Inter', system-ui, -apple-system, sans-serif;
--font-mono: 'JetBrains Mono', 'Fira Code', 'Cascadia Code',
  monospace;

```

\textbfSpace Grotesk All headings, UI labels, navigation, CTA buttons. Its slightly geometric, wide letterform reads as scientific without being cold.

\textbfInter All body copy, descriptions, abstracts. Maximally readable at small sizes.

\textbfJetBrains Mono All data values, IDs, tick counts, gene values, config params, code blocks. Never use a proportional font for numbers that align vertically.

3.2 Type Scale

```

--text-2xs: 0.625rem; /* 10px data labels, legend text */
--text-xs: 0.75rem; /* 12px captions, secondary metadata */
--text-sm: 0.875rem; /* 14px secondary body, UI labels */
--text-base: 1rem; /* 16px primary body copy */
--text-lg: 1.125rem; /* 18px lead text, card descriptions */
--text-xl: 1.25rem; /* 20px card titles, section labels */

```

```
--text-2xl: 1.5rem; /* 24px page section headers */
--text-3xl: 1.875rem; /* 30px page titles */
--text-4xl: 2.25rem; /* 36px major section heroes */
--text-5xl: 3rem; /* 48px landing hero text */
--text-6xl: 3.75rem; /* 60px ultra-large display (sparingly) */
```

3.3 Font Weight Usage

Use Case	Font	Weight
Page title	Space Grotesk	700
Section heading (h2)	Space Grotesk	600
Card title	Space Grotesk	600
Navigation links	Space Grotesk	500
Body copy	Inter	400
Secondary body	Inter	400
Lead paragraph	Inter	400 (but text-lg)
Button label	Space Grotesk	600
Metric value	JetBrains Mono	500
Data label	Inter	500
Code/config	JetBrains Mono	400
Phase badge	Space Grotesk	700

3.4 Line Heights

```
--leading-none: 1;
--leading-tight: 1.2; /* Headlines */
--leading-snug: 1.35; /* Card titles */
--leading-normal: 1.5; /* UI elements */
--leading-relaxed: 1.7; /* Body copy */
--leading-loose: 2.0; /* Spacious reading (monograph) */
```

3.5 Letter Spacing

```
--tracking-tighter: -0.03em; /* Large display headings */
--tracking-tight: -0.01em; /* Regular headings */
--tracking-normal: 0;
--tracking-wide: 0.04em; /* UI labels, phase badges */
--tracking-wider: 0.08em; /* All-caps micro-labels */
--tracking-widest: 0.14em; /* Section category labels (e.g. "
  EXPERIMENT") */
```

3.6 Prose Styles (Research Documents)

For the monograph, architecture doc, lab notebook, and thesis pages:

```
.prose {
  font-family: var(--font-body);
  font-size: var(--text-base);
  line-height: var(--leading-loose);
  color: var(--text-primary);
  max-width: var(--content-width); /* 800px */
```

```

h1 { font: 700 var(--text-4xl)/var(--leading-tight) var(--font-
  display); margin-bottom: 2rem; }
h2 { font: 600 var(--text-2xl)/var(--leading-tight) var(--font-
  display); margin-top: 3rem; margin-bottom: 1rem; }
h3 { font: 600 var(--text-xl)/var(--leading-snug) var(--font-display)
  ; margin-top: 2rem; }

p    { margin-bottom: 1.5rem; }
code { font-family: var(--font-mono); font-size: 0.9em; background:
  var(--bg-elevated); padding: 0.1em 0.4em; border-radius: var(--
  radius-sm); }
pre  { background: var(--bg-elevated); border: 1px solid var(--border
  -default); border-radius: var(--radius-md); padding: 1.5rem;
  overflow-x: auto; }

blockquote {
  border-left: 3px solid var(--accent-700);
  padding-left: 1.5rem;
  color: var(--text-secondary);
  font-style: italic;
}

table { width: 100%; border-collapse: collapse; }
th, td { padding: 0.75rem 1rem; border-bottom: 1px solid var(--border
  -subtle); text-align: left; }
th { color: var(--text-secondary); font: 500 var(--text-sm)/1 var(--
  font-body); letter-spacing: var(--tracking-wider); text-transform:
  uppercase; }
}

```

Part IV Spacing & Layout

4.1 Spacing Scale (4px base unit)

```

--space-0:    0;
--space-px:   1px;
--space-0-5:  0.125rem; /* 2px */
--space-1:    0.25rem;  /* 4px */
--space-2:    0.5rem;    /* 8px */
--space-3:    0.75rem;   /* 12px */
--space-4:    1rem;      /* 16px */
--space-5:    1.25rem;   /* 20px */
--space-6:    1.5rem;    /* 24px */
--space-7:    1.75rem;   /* 28px */
--space-8:    2rem;      /* 32px */
--space-10:   2.5rem;    /* 40px */
--space-12:   3rem;      /* 48px */
--space-14:   3.5rem;    /* 56px */
--space-16:   4rem;      /* 64px */
--space-20:   5rem;      /* 80px */
--space-24:   6rem;      /* 96px */
--space-32:   8rem;      /* 128px */

```

4.2 Border Radius

```
--radius-none: 0;
--radius-sm:   3px;
--radius-md:   6px;
--radius-lg:   10px;
--radius-xl:   14px;
--radius-2xl:  20px;
--radius-3xl:  28px;
--radius-full: 9999px; /* Pills, fully circular */
```

4.3 Layout Grid

```
--max-width:      1280px; /* Page container */
--content-width:   800px; /* Prose content */
--sidebar-width:   240px; /* Experiment detail sidebar */
--grid-gap:        1.5rem; /* 24px      standard grid gap */
--grid-gap-sm:     1rem;   /* 16px      tight grids */
--grid-gap-lg:     2rem;   /* 32px      loose grids */
```

4.4 Responsive Breakpoints

```
/* Mobile-first approach */
--bp-xs: 360px; /* Small phones */
--bp-sm: 640px; /* Large phones, small tablets */
--bp-md: 768px; /* Tablets */
--bp-lg: 1024px; /* Small laptops */
--bp-xl: 1280px; /* Desktops */
--bp-2xl: 1536px; /* Large screens */
```

4.5 Mobile Rules

- Sidebar becomes bottom tab bar (horizontal scroll) on $< \text{--bp-lg}$
- Replay canvas fills full viewport width on mobile
- World map uses touch pan/pinch instead of mouse
- Timeline events stack full-width
- Metric cards: 2 columns on mobile, 4 on desktop
- Archive grid: 1 column on mobile, 2 on tablet, 3 on desktop

Part V Effects & Surfaces

5.1 Glass Morphism

Used on: cards, sidebars, modals, tooltip panels, the agent inspector.

```
.glass {
  background: hsl(222, 24%, 10%, 0.75);
  backdrop-filter: blur(20px) saturate(160%);
  -webkit-backdrop-filter: blur(20px) saturate(160%);
  border: 1px solid hsl(222, 20%, 22%);
  border-top-color: hsl(222, 20%, 28%); /* Subtle top highlight */
}

.glass-strong {
  background: hsl(222, 24%, 8%, 0.88);
  backdrop-filter: blur(32px) saturate(180%);
}
```

5.2 Shadows

```
--shadow-none: none;
--shadow-sm:    0 1px 2px hsl(222, 40%, 2%, 0.40),
                0 1px 4px hsl(222, 40%, 2%, 0.20);
--shadow-md:    0 4px 8px hsl(222, 40%, 2%, 0.45),
                0 2px 16px hsl(222, 40%, 2%, 0.25);
--shadow-lg:    0 8px 16px hsl(222, 40%, 2%, 0.50),
                0 4px 32px hsl(222, 40%, 2%, 0.30);
--shadow-xl:    0 16px 32px hsl(222, 40%, 2%, 0.55),
                0 8px 64px hsl(222, 40%, 2%, 0.35);

/* Accent glow shadows */
--shadow-accent-sm: 0 0 12px hsl(172, 80%, 48%, 0.18);
--shadow-accent-md: 0 0 24px hsl(172, 80%, 48%, 0.22);
--shadow-accent-lg: 0 0 48px hsl(172, 80%, 48%, 0.28);
```

5.3 Elevation System

Elevation	Surface	Shadow	Blur	Use Case
0	\texttt{bg-void}	none	none	Page background
1	\texttt{bg-primary}	\texttt{shadow-sm}	none	Main content area
2	\texttt{bg-secondary}	\texttt{shadow-md}	none	Cards, panels
3	\texttt{bg-surface}	\texttt{shadow-lg}	glass	Dropdowns, tooltips
4	\texttt{bg-elevated}	\texttt{shadow-xl}	glass-strong	Modals, overlays
5	Accent	\texttt{shadow-accent-md}	none	Featured, active, selected

Part VI Animation System

6.1 Easing Functions

```
--ease-linear:    linear;
--ease-in:        cubic-bezier(0.4, 0, 1, 1);
--ease-out:       cubic-bezier(0, 0, 0.2, 1);
--ease-in-out:    cubic-bezier(0.4, 0, 0.2, 1);

/* Genesis-specific easings */
```

```
--ease-out-quart:    cubic-bezier(0.25, 1, 0.5, 1);        /* Smooth,
    confident */
--ease-spring:       cubic-bezier(0.34, 1.56, 0.64, 1);     /* Springy
    microinteractions */
--ease-overshoot:    cubic-bezier(0.68, -0.55, 0.27, 1.55); /* Playful
    overshoot */
--ease-slow-out:     cubic-bezier(0.16, 1, 0.3, 1);         /* Dramatic
    reveal */
```

6.2 Duration Scale

```
--duration-instant:  50ms;      /* State flashes, immediate feedback */
--duration-fast:      100ms;     /* Hover states */
--duration-normal:    200ms;     /* Most transitions */
--duration-moderate:  300ms;     /* Panel slides, dropdown opens */
--duration-slow:      500ms;     /* Page section reveals */
--duration-slower:    700ms;     /* Hero elements, major transitions */
--duration-page:      1000ms;    /* Full-page transitions */
```

6.3 Micro-interaction Specifications

Every interactive element is specified here. No guessing.

`\textbfButton primary (teal)`

```
rest:      bg=accent-400, shadow=none, scale=1.0
hover:     bg=accent-300, shadow=accent-sm, scale=1.02 [150ms ease-out-
    quart]
active:    bg=accent-600, shadow=none, scale=0.97 [100ms ease-in]
focus:    ring: 2px accent-400 offset 2px
```

`\textbfButton ghost`

```
rest:      bg=transparent, border=border-default, color=text-secondary
hover:     bg=bg-surface, border=border-hover, color=text-primary [150ms
    ease-out]
active:    bg=bg-elevated [100ms ease-in]
```

`\textbfExperiment Card`

```
rest:      bg=bg-secondary, border=border-subtle, shadow=shadow-sm,
    translateY=0
hover:     bg=bg-surface, border=border-hover, shadow=shadow-md,
    translateY=-3px [200ms ease-out-quart]
    thumbnail: scale=1.04 (inner only, parent overflow:hidden)
                [400ms ease-out-quart]
    world preset badge: color transitions to accent
                [150ms ease-out]
active:    translateY=-1px, shadow=shadow-sm [100ms ease-in]
featured:  border-top: 2px solid accent-400 (always on)
```

`\textbfTimeline Event Row`

```
rest:      opacity=1, pl=var(--space-4)
hover:     bg=hsl(222,20%,14%,0.5), pl=var(--space-5) (left nudge) [150
    ms ease-out]
    event dot: scale=1.3, glow appears [200ms ease-spring]
active:    bg=bg-surface
```

\textbf{Tab (sidebar navigation)}

```
rest:      color=text-secondary, bg=transparent
hover:     color=text-primary, bg=hsl(222,20%,14%) [100ms ease-out]
active:    color=accent-400, bg=accent-900
           left border: 2px solid accent-400
```

\textbf{Layer Toggle Chip (world map)}

```
rest:      bg=bg-elevated, color=text-secondary, border=border-default
hover:     bg=bg-surface, color=text-primary [150ms ease-out]
active:    bg=accent-900, color=accent-300, border=accent-700
toggle:    layer fades in/out on canvas: 200ms ease-out
```

\textbf{World Map Agent Dot}

```
render:    radius=3px baseline; filled with colony color
           opacity = agent.health / 100 (full health = fully opaque)
move:      lerp between tick positions (linear, per-frame)
hover:     radius expands to 6px [150ms ease-spring]
           inspector panel slides in from right [300ms ease-out-quart]
death:     dot flashes white (50ms), then red (200ms), then fades to 0
           opacity (300ms)
birth:     dot appears at parent midpoint, scale 0 1 [300ms ease-spring]
```

\textbf{Metric Counter (on viewport enter)}

```
trigger: IntersectionObserver, 0.1 threshold
effect:   value counts from 0 to final value
easing:   ease-out-quart applied to the count interpolation
duration: 800ms
```

\textbf{Search Bar}

```
rest:      width=200px, bg=bg-secondary, border=border-default
focus:    width=320px (animated), border=border-accent, shadow=accent-sm
           [250ms ease-out-quart]
results:   slide in from below, opacity 0 1 [200ms ease-out]
```

\textbf{Modal}

```
open:      backdrop fades in (bg:hsl(222,40%,2%,0.7)) [250ms ease-out]
           panel: scale 0.95 1 .0, opacity 0 1 [300ms ease-out-quart]
close:     reverse, both [200ms ease-in]
```

\textbf{Toast Notification}

```
enter:     slide up from bottom-right (translateY: 80 p x 0 ), opacity 0
           1 [300ms ease-spring]
auto-exit: after 4000ms, slide down + fade [200ms ease-in]
hover:     auto-exit timer pauses
```

\textbf{Replay Controls}

```
play/pause: icon morphs with SVG path animation [200ms ease-in-out]
speed:      click cycles through 0.5 124 with badge update
           [100ms]
tick counter: monospace, updates each rendered frame
scrubber:   drag to seek canvas re-renders at target tick on mouseup
```

6.4 Page-Level Transitions

```
Route change:
  outgoing: opacity 1 0 , translateY 0 (-8px) [200ms ease-in]
  incoming: opacity 0 1 , translateY (8px) 0 [300ms ease-out]
  total: 500ms (overlap at 200ms)

Section reveals (scroll):
  trigger: IntersectionObserver, threshold 0.15
  effect: opacity 0 1 , translateY 20 p x 0 [500ms ease-slow-out]
  stagger: 80ms between sibling elements
```

6.5 Canvas Animation Replay Renderer

```
// Per-frame rendering contract
function renderFrame(tick, agents, worldLayers, config) {
  // 1. Clear canvas
  ctx.clearRect(0, 0, canvas.width, canvas.height);

  // 2. Render base layer (biome map pre-rendered ImageData, don't
  //    recompute)
  ctx.putImageData(biomeCacheImageData, 0, 0);

  // 3. Render optional overlay layers (rivers, resources, habitability
  //    )
  if (layers.rivers) renderRiverOverlay(ctx, worldState);
  if (layers.habitability) renderHeatmap(ctx, worldState.habitability,
    0.4);

  // 4. Render death density (persistent death heatmap)
  if (layers.deaths) renderHeatmap(ctx, worldState.death_density, 0.35)
  ;

  // 5. Render agents
  for (const agent of agents) {
    const lerped = lerpPosition(agent.prevPos, agent.pos, frameProgress
    );
    const alpha = Math.max(0.2, agent.health / 100);
    const radius = agent.hovered ? 6 : 3;
    const color = COLONY_COLORS[agent.colony_id];

    ctx.beginPath();
    ctx.arc(lerped.x, lerped.y, radius, 0, Math.PI * 2);
    ctx.fillStyle = `rgba(${color.r}, ${color.g}, ${color.b}, ${alpha})`
    ;
    ctx.fill();

    // Death flash animation
    if (agent.deathFlash > 0) {
      ctx.fillStyle = `rgba(255, 255, 255, ${agent.deathFlash})`;
      ctx.fill();
      agent.deathFlash = Math.max(0, agent.deathFlash - 0.08);
    }
  }

  // 6. Render hovered agent highlight ring
  if (hoveredAgent) {
```

```

    ctx.strokeStyle = 'rgba(255, 255, 255, 0.8)';
    ctx.lineWidth = 1.5;
    ctx.beginPath();
    ctx.arc(hoveredAgent.x, hoveredAgent.y, 8, 0, Math.PI * 2);
    ctx.stroke();
  }
}

```

Part VII Icon Vocabulary

Genesis uses a minimal, consistent icon set. No icon packs. Custom SVGs where needed.

7.1 Event Type Icons

Event	Icon	Color
Birth	\texttt (expanding ring)	\texttt-event-birth
Death	\texttt (cross)	\texttt-event-death
Milestone	\texttt (diamond)	\texttt-event-milestone
Disaster	\texttt (triangle, storm)	\texttt-event-disaster
Dispute	\texttt (crossed swords, custom SVG)	\texttt-event-dispute
Climate Epoch	\texttt\textasciitilde (wave)	\texttt-event-climate
Extinction	\texttt (solid square, black)	\texttt-event-extinction

7.2 Navigation Icons

Page	Icon
Archive	Grid (22)
Observatory	Circle with dot (eye symbol)
Replay	Triangle (play)
Chronicle	Vertical line with ticks
World	Globe outline
Lineage	Tree branch
Genome	Helix (simplified)
Conditions	Sliders
Research	Book open
Timeline	Horizontal line with dots
Origin	Sparkle / star

7.3 Icon Rules

- Size: 16px for inline, 20px for navigation, 24px for feature icons
- Stroke: 1.5px, round linecap, round linejoin
- Never fill + stroke together (pick one)
- Never use icon fonts always SVG

Part VIII Component Visual Specifications

8.1 Navbar

```
Height: 60px
Position: fixed top, full width
Background: hsl(222, 28%, 6%, 0.85) + blur(20px)
Border-bottom: 1px solid border-subtle (always present, not on-scroll)

Left:
  Genesis wordmark      Space Grotesk 700, 18px, accent-400
  Phase badge           pill, bg:accent-900, color:accent-400, font:mono 11px

Center:
  Nav links             Space Grotesk 500, 14px, text-secondary
  Hover: color text-primary + underline (2px accent-700 bottom)
  Active: color accent-400 + underline (2px accent-400)
  Gap between links: 32px

Right:
  GitHub icon           20px, text-tertiary      text-primary on hover
  "Command" button (admin only) teal pill button, 500 font, 14px
```

8.2 Experiment/Civilization Card

```
Width: flexible (grid column)
Border-radius: radius-xl (14px)
Border: 1px solid border-subtle
Background: bg-secondary

Thumbnail area:
  Height: 140px
  overflow: hidden
  Object: world.png, cover fit
  Overlay: linear-gradient(to bottom, transparent 60%, bg-secondary 100%)

Content area:
  Padding: space-5 (20px)

  ID row:
    font: mono 11px, text-tertiary, tracking-wide
    "EXP-20260627-XY81"

  Title:
    font: Space Grotesk 600, text-xl, text-primary
    max 2 lines, line-clamp

  Tags row:
    Pills: bg-elevated, border-subtle, font:Inter 12px, text-secondary
    Gap: space-2

  Stats row (mono):
    "18 agents      7,970 ticks (22 yr)"
    font: JetBrains Mono 12px, text-tertiary

  Meta row:
```

```
"Published Jul 13, 2026"  
font: Inter 12px, text-tertiary
```

```
Featured variant:  
border-top: 2px solid gradient-card-top  
shadow: shadow-accent-sm
```

8.3 Metric Card

```
Padding: space-6  
Border-radius: radius-lg  
Background: bg-secondary  
Border: 1px solid border-subtle  
  
Label:  
  font: Inter 500, text-xs, text-tertiary, tracking-wider, uppercase  
  
Value:  
  font: JetBrains Mono 500, text-3xl, text-primary  
  margin-top: space-2  
  
Unit (if any):  
  font: Inter 400, text-sm, text-secondary, inline after value  
  
Delta (optional):  
  font: Inter 500, text-sm  
  color: green if positive, red if negative  
  "    +12% vs average"  
  margin-top: space-1  
  
Accent variant:  
  border-left: 3px solid accent-700  
  
Warning variant:  
  border-left: 3px solid event-disaster  
  
Critical variant:  
  border-left: 3px solid event-death
```

8.4 Phase Badge (Timeline)

```
Display: inline-flex, align-center  
Padding: space-1 space-3  
Border-radius: radius-full  
font: Space Grotesk 700, text-xs, tracking-wider, uppercase  
  
Phase 0-1: bg=hs1(222,20%,20%), color=text-secondary  
Phase 2-4: bg=hs1(195,30%,18%), color=hs1(195,70%,65%)  
Phase 5-7: bg=hs1(172,30%,14%), color=accent-300  
Phase 8-9: bg=accent-900, color=accent-400  
Phase 10+: bg=hs1(280,30%,18%), color=hs1(280,60%,70%)
```

8.5 Code/Config Viewer

```

Background: bg-elevated
Border: 1px solid border-default
Border-radius: radius-md
Padding: space-6

Header bar:
  filename (mono, text-sm, text-secondary)
  Copy button (right-aligned)

Content:
  font: JetBrains Mono 400, text-sm, leading-relaxed
  JSON syntax highlighting:
    keys:      accent-300
    strings: hsl(45, 80%, 65%)    (amber)
    numbers: hsl(195, 75%, 65%)   (light blue)
    booleans: event-birth / event-death
    null:     text-tertiary

```

Part IX World Map Rendering Specification

9.1 Canvas Setup

```

Resolution: 1:1 cell-to-pixel mapping (512 512 for a 512-wide world)
Scaling: CSS transforms for zoom (not canvas resize avoid re-render on zoom)
Pan: transform-origin center, translate on drag
Cursor: crosshair default, grab on pan

```

9.2 Layer Rendering Order

1. Ocean fill (biome-ocean, flat)
2. Elevation gradient (if elevation layer active)
3. Biome colors (if biome layer active) per-cell fill
4. River overlay: blue strokes, width=1, opacity proportional to flow
5. Lake overlay: semi-transparent biome-lake fill
6. Resource heatmap (if resource layer active) color mapped per resource
7. Habitability heatmap (if habitability layer active) opacity 0.5
8. Death density heatmap (if deaths layer active) opacity 0.35
9. Colony territory outlines (thin dashed, colony color, convex hull)
10. Agent dots
11. Hovered cell highlight (1px white border around cell)
12. Selected agent ring

9.3 Hover Tooltip

```

Position: follows cursor (offset: +12px right, +8px below)
Overflow: flip to left if near right edge

Content:
  [Biome name]      hsl(biome_color)
  Elevation: 0.62

```

```
Rainfall: 823 mm/yr
Temperature: 12.4 C
Wood: 0.68
Stone: 0.31
Iron: 0.02
[Causal explainer text if available]
```

\textitEnd of GDS v1.0

\textbfDocument produced by Antigravity AI, July 2026 \textbfCompanion: GPS GES GCR